Predicting Bitcoin's Hourly Price Changes: The Direction Is Noise, the Size Is Not

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***Abstract*—We forecast the 24 hourly price changes of Bitcoin's next day from hourly Bitstamp data (2020–2026). An exploratory analysis on the training years shows that the direction of hourly moves is almost noise, while their size is persistent, follows the clock and drifts over time; these findings shape the features, targets and models. With chronological blocks, choices made without the test block and comparison against simple rules, no model (linear, trees, neural networks or the zero-shot foundation model Timestral) beats a forecast of "no change" for the returns, apart from a small reversal one hour ahead (AUC 0.54). The size of each hourly move, however, is predictable: monthly-refitted Gradient Boosting reaches $R^2 = 0.148$, and its average with Timestral 0.164, against 0.072 for the best simple rule; recalibrated 80% ranges keep 80% coverage, and next-day volatility reaches $R^2 = 0.51$ against 0.09. In a trading simulation these forecasts do not earn more money but make trading safer. Code: https://github.com/Descalabreitor/IAI600-final-project.**

***Index Terms*—Bitcoin, time-series forecasting, volatility, gradient boosting, conformal prediction, LSTM, foundation models**

I. Introduction

The assigned task is to predict, for each of the 24 hours of the next day, how much the price of Bitcoin will change, $r_h = \mathrm{Close}_h / \mathrm{Close}_{h-1} - 1$, from the hours before. We make this forecast from every hour, for the 24 hours that follow; the next day is the case of the forecast made at 23:00. Such short-horizon forecasts matter for risk management, position sizing and the timing of trades. Bitcoin is a demanding case: it trades 24/7, its returns are heavy-tailed, and its behaviour shifts between bull and bear markets and after events such as the launch of the US spot ETFs in January 2024.

Two classical results frame the problem. In an efficient market, past prices should not predict the direction of future returns [1]; the size of the moves, on the other hand, clusters in time and is predictable [2], [3]. Financial time series also make it easy to fool oneself: overlapping windows and random splits leak the future into the training data [4]. We therefore let a data analysis decide which questions are answerable, evaluate every model against a simple rule on a test period never used for choices, and finally ask what the forecasts are worth to a trader.

Related work. Machine-learning studies of Bitcoin report direction accuracies of only 52–56%, for daily moves [5] and for horizons of 1 to 60 minutes [6], and the trading profits of the latter vanish after transaction costs. Volatility, by contrast, is usually forecast well from its own recent averages over several horizons [7]. Deep forecasters built on MLP, recurrent, convolutional and Transformer backbones are reviewed and benchmarked in [8], which points to pretrained foundation models as the next step: Chronos [9], TimesFM [10] and Moirai [11] forecast unseen series zero-shot, and Timestral [12] belongs to this line. We compare all three families on the same data and blocks, for the direction and for the size of the moves.

II. Methodology

A. Data and Preparation

The data is the Kaggle "Bitcoin Historical Data" set [13] of 1-minute Bitstamp BTC/USD candles, pinned to one version and aggregated to hourly candles without dropping missing rows, so the hourly grid stays continuous. The minute file hides its gaps as flat, zero-volume candles at the previous close: 95% of the minutes of 2012 and 25–40% of 2013–2016, but 1–4% per year since 2018. Outage hours are flagged, not removed, and we use the 59,331 hours from 1 January 2020 to October 2026, so that training is close to the current market. Extra inputs were tested: Binance BTC and ETH candles with order flow [14], about 40 macro series from FRED [15] and Yahoo Finance, and the times of US CPI, payroll and FOMC releases.

B. Exploratory Data Analysis

Every analysis that could steer a model is fitted on the training block and checked on validation. Fig. 1 shows the central fact: returns are almost uncorrelated (about -0.03 at lags 1, 2 and 24; an AR(24) model does worse than the mean on validation), while their absolute values are strongly autocorrelated (0.30 at lag 1, still 0.10 a week later). The direction is almost noise; the size is not. The size also follows the clock, and this rhythm drifts (Fig. 2): the 14 UTC peak of the US session grows from 2022 and jumps after the spot ETFs, weekends become calmer, and the average hourly move falls from 0.50% in training to about 0.30% later. US releases move Bitcoin 2–3 times more than the same hour on other days, whereas ETH and the macro variables move together

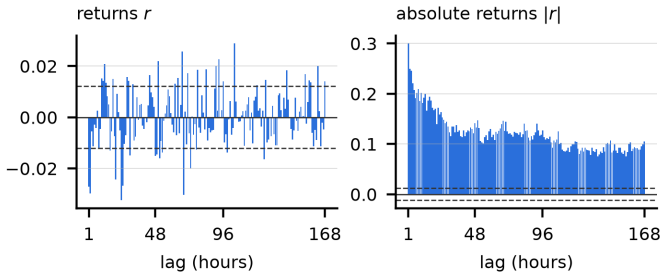


Fig. 1. Autocorrelation of hourly returns (left) and of their absolute value (right), lags 1–168 h, training block 2020–2022. Dashed lines: 95% band of white noise.

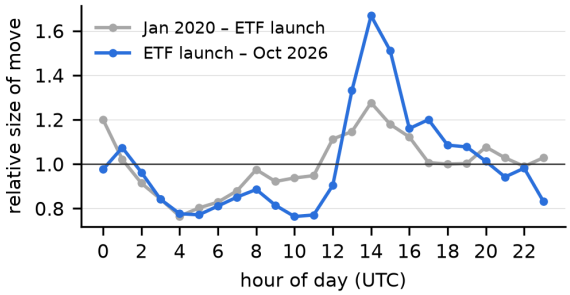


Fig. 2. Size of the hourly move by hour of day (UTC), relative to the average hour, before and after the launch of the US spot ETFs.

with Bitcoin but never lead it. Table I lists how each finding shaped the modelling.

C. Targets and Features

A forecast is made at every hour (the *origin*) with 30 days of history and 24 target hours. Besides the 24 returns we forecast (a) the direction of the next hour, (b) the size $|r|$ of each of the next 24 hours, (c) an 80% range for each of them and (d) the next day's return, direction and volatility. Features are lagged returns, realised volatilities over 24–168 hours, high–low ranges, volume ratios and the calendar. In the *long format* each origin gives 24 rows, one per predicted hour, with columns for that hour (its average size, its size a day and a week earlier): one model learns the common structure ("volatile weeks make every hour volatile") from 24 times more rows.

D. Models

Classical: Ridge [16], Lasso [17], ElasticNet [18], k-nearest neighbours, SVR/SVC, Random Forest [19], Extra Trees [20] and Gradient Boosting (GB) [21], alone and averaged. Ranges come from GB quantile regression [22] (10% and 90%), recalibrated by *rolling conformal calibration* [23], [24]: the half-width is rescaled so that the last 30 days of known outcomes would have had 80% coverage; the *Mondrian* variant does this separately for release hours. Volatility models are also refitted every month on all data known by then. *Neural:* MLPs and LSTMs [25] reading a 168-hour window normalised by its own statistics, the stationarisation step common to deep

TABLE I
FROM THE DATA ANALYSIS TO THE MODELLING CHOICES.

Finding	Consequence
Price rose 30-fold; 72% of test hours above the training range	forecast changes; scale-free features only
Returns almost uncorrelated	"no change" as baseline; strong regularisation; add targets the data can answer (size, ranges, next hour)
Size persistent and tied to the clock	volatility and calendar features; one row per predicted hour; trees
Heavy tails (excess kurtosis 55)	quantile ranges; MAE skill; Huber loss for networks
Rhythm and level drift	data from 2020; monthly refits; rolling calibration of ranges
Release hours move 2–3×	release flags; ranges calibrated apart for them
ETH, macro do not lead	tested only as optional inputs

TABLE II
TUNED GRIDS AND SETTINGS CHOSEN ON VALIDATION.

Model (target)	Grid → chosen
Ridge (returns)	$\alpha \in 0.1$–$10^9 \to 10^5$ (shrinks almost to the zero forecast)
GB (hourly $\lvert r\rvert$)	learning rate $\{0.05, 0.1\}$, depth $\{2, 3, 6\} \to 0.1$, 6; 1,000 rows per leaf
GB (next-day vol.)	learning rate 0.05, depth 2, 200 trees
Monthly refit	expanding, 24-month window, half-life 6/12 months → 12 months ($\lvert r\rvert$), expanding (daily)
LSTM ($\lvert r\rvert$)	2 layers, hidden $\{32, 64, 128\} \to 64$, dropout 0.2, 5 seeds
Timestral	28M/209M → 28M; series $\lvert r\rvert$; 180-day context; clock, release calendar, range and volume

forecasters [8]; the best LSTM also sees the clock and release flags of each predicted hour, weights the week by traded volume (attention) and is fine-tuned monthly. *Foundation model:* Timestral [12], pretrained by Mistral AI on mostly synthetic series (real ones enter only in a second training stage), returns 21 quantiles of each of the next 24 hours. It is used zero-shot: we only choose on validation what it reads (series, context length, covariates) and its size. Table II lists the chosen settings.

III. Experimental Setup

A. Tools

All code is Python 3.12 with NumPy, pandas, SciPy, scikit-learn [26], PyTorch [27] and the `timestral` package [12], in six Jupyter notebooks (base data, data analysis, classical models, neural networks, Timestral, trading) sharing one `utils` package. Code: https://github.com/Descalabreitor/IAI600-final-project.

B. Data Split and Model Selection

Blocks are cut by the date of the predicted hours: *train* 2020–2022, *validation* Jan 2023 – Jun 2024 and *test* Jul 2024 – Oct 2026. A 192-hour purge before each border keeps every target hour in one block only [4]. Every choice (model,

hyperparameters, features, refit schedule, calibration) is made on train and validation; the chosen model is refitted on both and scored once on test. Test scores use every hourly origin; the 829 forecasts made at 23:00, which share no target hour, give the daily targets and every significance test. Monthly refits and rolling calibration only use outcomes already known, as a forecaster in real time would. Hourly targets are tuned on validation, daily targets on four walk-forward folds. Models are chosen by the metric they are judged on: a model trained for the mean but chosen by MAE drifts towards the zero forecast. Networks stop early on the last six months of train, never on a random hold-out.

C. Metrics and Baselines

Returns are scored by the MAE skill against the zero forecast ($1 - \mathrm{MAE}/\mathrm{MAE}_0$) and the information coefficient (IC, correlation of forecast and outcome); direction by the AUC; size and volatility by R^2; ranges by coverage and width. The R^2 of $|r|$ has a low ceiling, since one hour's $|r|$ is a very noisy measure of the volatility behind it [28], so 0.1–0.2 is already real skill; squared error still ranks forecasts correctly against such a proxy [29]. Baselines are the zero forecast, the reversal of the last hour, last week's volatility times the hour profile (of the training block or of the last 90 days), and "always up". Differences are tested with a moving-block bootstrap [30] over the 829 test days and Diebold–Mariano tests [31]; a difference is *reliable* when its 95% interval excludes zero.

D. Trading Simulation

$10,000 is traded spot, long only above the 50-day mean, with fixed, untuned rules. Each day the position is sized so that a bad day (the 10% worst case, from the next-day volatility forecast) costs 1.5% of the equity, with a stop and fees of 0.10% per side. Four flows differ only in that forecast: last week's volatility, monthly GB, Timestral and their average. We compare money, the Sharpe ratio (mean over standard deviation of daily returns, compared by a paired block bootstrap) and drawdowns (falls from the previous equity peak); the chance of a drawdown above 20% within a year comes from 5,000 years resampled in 20-day blocks. As a stress test, futures leverage is set daily from each forecast at the same average leverage, high enough for liquidations (a fall of about $1/k$ below the entry at leverage k, measured on hourly lows) to occur.

IV. Results and Discussion

Table III summarises the test results.

A. Returns and Direction

No model beats "no change" for the hourly returns: every tuned model ends within rounding of the zero forecast's MAE, and more model freedom always meant worse validation scores; the chosen Ridge shrinks almost to zero. Selecting by squared error instead of MAE leaves a tiny ranking signal (IC 0.02, interval excluding zero) that is far too weak to beat

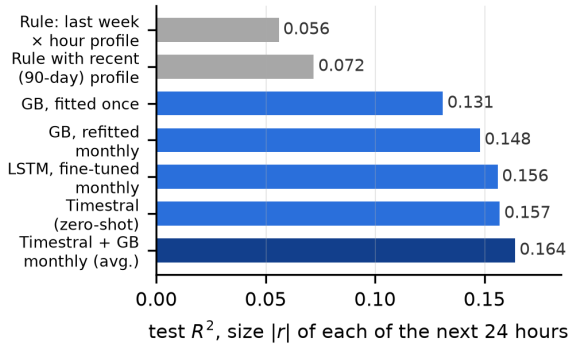


Fig. 3. Test R^2 for the size of each of the next 24 hourly moves: simple rules (grey), single models (blue) and the chosen average (dark blue).

zero, as market efficiency predicts [1]. The one exception is the next hour, which tends to reverse the last one: the simple rule reaches AUC 0.528, GB with Binance order flow 0.538. The effect is small, before trading costs, and gone from the second hour on. The weak lag-1 autocorrelation of Fig. 1 does not contradict it: the autocorrelation is dominated by a few huge hours, whereas the AUC counts correct signs.

B. Size of the Hourly Moves

The size is where the models have clear skill (Fig. 3). The long format was the largest modelling gain (validation R^2 0.052 to 0.097). On test, GB fitted once explains 13.1% of the variance of each hourly $|r|$, almost twice the best simple rule (0.072). Refitting every month raises it to 14.8% ($+0.017$ $[+0.010, +0.024]$): in a drifting market, keeping the model current is worth more than any extra input. None of the candidate features from the analysis, nor Binance and ETH, improved the trees, because recent volatility and the clock already carry their information. Release hours stay under-predicted (forecast about $0.65\times$ their actual size): with a few hundred training rows they are fewer than one leaf of the tuned GB.

Networks needed the same knowledge to compete: an LSTM gains $+0.031$ when told the clock and release flags of the hour it predicts (what the per-hour columns do for the trees), and with volume attention and monthly fine-tuning reaches 0.156, narrowly above GB ($+0.008$ $[+0.001, +0.015]$). Timestral, never trained on our data, matches it (0.157), provided it reads $|r|$ rather than r, a long (180-day) context and the release calendar; the 209M model added almost nothing over the 28M one. The plain average of Timestral and GB is the best result (0.164, $+0.016$ $[+0.013, +0.020]$ over GB). Feeding Timestral's forecast into GB as a feature was worse, and model families converge within 0.01 R^2: averaging two different families is what helps.

C. Prediction Ranges and Next-Day Volatility

The GB quantile ranges are 28% narrower than a fixed range and adapt to calm and wild markets, but cover only 78% of the test hours, because the test years differ from the training

TABLE III
TEST-BLOCK RESULTS (JUL 2024 – OCT 2026) AGAINST THE BEST SIMPLE RULE. MODELS WERE CHOSEN ON TRAIN AND VALIDATION; THE TRADING RULES WERE FIXED IN ADVANCE. BRACKETS: 95% BLOCK-BOOTSTRAP INTERVAL.

Question	Model	Test result	Simple rule	Beats rule?
24 hourly returns (assigned task)	linear, KNN, trees, MLP, LSTM, Timestral	MAE skill ≤ 0 for all (best -0.0002); IC ≤ 0.021	zero forecast	no
Direction of the next hour	GB classifier	AUC 0.538; 56% right on the 10% most confident calls	last-hour reversal: AUC 0.528	yes, small
Size $\lvert r\rvert$ of each next hour	Timestral + GB refitted monthly	$R^2 = \mathbf{0.164}$ (Timestral 0.157, LSTM 0.156, GB 0.148)	last week $\times$ recent (90-day) hour profile: 0.072	yes
80% range for each next hour	GB quantile + rolling Mondrian conformal	coverage 80.0% (calm 78%, wild 82%, release hours 82%), width 0.98%	fixed range: 87.6%, 1.28%	yes
Next-day return / direction	Ridge / Random Forest	IC 0.051 / AUC 0.509	zero / always up	no
Next-day volatility	tree ensemble; GB refitted monthly	$R^2 = 0.510$ [0.45, 0.56]; 0.517	last week's mean: 0.094	yes
Trading \$10,000	GB + Timestral risk forecast	same money as naive; liquidations 3 vs. 12; P(drawdown $> 20\%$) 21% vs. 27%	last week's volatility	safety only

years. A correction measured once on validation does not fix this; rolling conformal calibration, which keeps re-measuring the miss rate, restores 80%. Release hours, covered only 56% of the time, reach 82% when calibrated apart (Mondrian). Timestral's own quantiles are calibrated without any correction (80.3%) and, thanks to the release calendar, cover 70% of release hours. At the daily scale, volatility is predictable as well: a tree ensemble explains 51% of the next day's log volatility against 9% for last week's average, and a monthly refit again helps (0.517).

D. Trading Simulation

As risk meters, the model forecasts are far better than last week's volatility, but they do not earn more money: all four flows end close together, the naive one slightly ahead, and no Sharpe difference is reliable. A risk-sized long position can only grow or shrink, never take the right side; in the bull market of validation, cutting it before volatile days even cost money. The forecasts do make trading safer, because they know *when* risk is high: at the same average leverage ($\approx 13\times$) the leverage lands on calm days, and the averaged forecast is liquidated 3 times on test instead of 12. The chance of a drawdown above 20% within a year falls from 27% to 21%, and to 10–12% with a forecast-shaped intraday stop and a weekly loss limit, against 54% for the same trend rule without risk control. Timestral does not predict the direction either: trading on its expected return lost money.

V. CONCLUSION

The analysis of the data decided most of the project: it showed that hourly returns are close to a random walk, so that none of the models we tried on the assigned target, from linear models to a pretrained foundation model, beats the zero forecast, and that the size of the moves is persistent, clock-driven and drifting, which pointed to scale-free volatility features, one row per predicted hour, monthly refits and rolling calibration. We therefore recommend answering the task with the expected size and an 80% range for each hour, for which the average of Timestral and monthly-refitted Gradient Boosting explains 2.3 times the variance of the best simple rule. These forecasts control trading losses, not returns.

We learned that the evaluation design matters more than the model: chronological blocks with a purge, choices made without the test block, simple baselines and bootstrap intervals separate real signal from noise. Recency beat data volume and extra sources, and a pretrained model, used as it comes, matched careful feature engineering where there was signal but could not create signal where there was none.

The study has one test period of 2.25 years, many tests on the returns without a multiple-testing correction, release hours that remain hard to forecast, and a trading simulation without slippage or funding costs. Timestral's second training stage used real series that are not listed; if they include Bitcoin after mid-2024, its test scores may be optimistic. Future work includes a rolling-origin evaluation over 2023–2026, a cost-aware test of the next-hour signal, a forward test of Timestral, other deep backbones [8], and options, where the size of a move pays without its direction.

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Contributions. Alberto Prats: classical models and hyperparameter tuning; Adrián Perera: data pipeline, unification of the notebooks, neural networks and Timestral; Filip Botlo: Binance features, monthly refits, conformal ranges, course-method pipeline, trading simulation; Adrián Muñiz: exploratory data analysis and macro correlations. We thank the author of the Kaggle dataset, Binance, FRED and Mistral AI for the public data and models, and the course staff.

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